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Cartridge and applicator for drug delivery and microneedle patch used therefor

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Abstract

Proposed is a microneedle patch including a body in which at least one opening is formed, and a microneedle body which is disposed in the opening and has at least one microneedle formed therein, wherein when an external force is applied in a first direction, the microneedle body moves in the first direction so as to administer a drug in the microneedle to a user's skin.

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Background/Summary

CROSS REFERENCE TO RELATED APPLICATIONS (1) This is a continuation of U.S. patent application Ser. No. 17/633,474 filed on Feb. 7, 2022, which is a national phase of International Patent Application No. PCT/KR2021/002145, filed on Feb. 19, 2021, which claims priority to U.S. Provisional Application No. 62/979,224 filed on Feb. 20, 2020, and Korean patent application No. 10-2020-0116801 filed on Sep. 11, 2020, contents of each of which are incorporated herein by reference in their entireties.

TECHNICAL FIELD
(1) The present disclosure relates to an applicator for drug delivery and a microneedle patch used therefor.
BACKGROUND ART
(2) The conventional drug injection method using a syringe has problems such as the need for a healthcare professional to administer an injection and the pain accompanying the injection. To solve these problems, a microneedle type drug injection system was developed.
(3) The microneedle is a system that delivers active ingredients into the skin through the stratum corneum, which is the skin barrier layer. It is a new system that combines the efficacy of a conventional syringe with the convenience of a patch and has made many technological advances in recent years.
(4) Yet, in the case of the patch-type microneedle, there is a problem in that a user himself or herself has to attach the microneedle to the skin and inject the drug, so that the user's active attachment action is still required.
DISCLOSURE
Technical Problem
(5) Accordingly, the present disclosure is proposed to solve the above-described problems and an objective of the present disclosure is to provide an applicator in the form of a wearable device that can automatically inject drugs using a microneedle patch.
(6) Another objective of the present disclosure is to provide a microneedle patch that can be used multiple times and can be used in replaceable cartridges in combination with a wearable device.
Technical Solution
(7) According to an embodiment of the present disclosure for achieving the objectives as described above, provided is a microneedle patch including a body in which at least one opening is formed,

and a microneedle body which is disposed in the opening and has at least one microneedle formed therein, wherein when an external force is applied in a first direction, the microneedle body moves in the first direction so as to administer a drug in the microneedle to a user's skin.

Advantageous Effects

(8) According to an embodiment of the present disclosure, the microneedle patch can be used multiple times by rotating since it moves to administer the drug when an external force is applied thereto and moves back to the original position thereof, and it is configured in a form that can be used in combination with a wearable device so that the drug can be injected in a way convenient for users.

Description

DESCRIPTION OF DRAWINGS

(1) FIG. **1** illustrates the overall appearance of a drug delivery device according to an embodiment.
(2) FIG. **2** is a half exploded view illustrating the general configuration of a drug delivery device according to an embodiment.
(3) FIG. **3** is a front stereoscopic view of a microneedle patch according to the first embodiment.
(4) FIG. **4** is a rear stereoscopic view of a microneedle patch according to the first embodiment.
(5) FIG. **5** is a cross-sectional view of a microneedle patch according to the first embodiment.
(6) FIG. **6** is a bottom view of a microneedle patch according to the first embodiment.
(7) FIG. **7** is a bottom perspective view illustrating a patch assembly according to the second embodiment.
(8) FIG. **8** is a top view illustrating a microneedle patch according to the third embodiment.
(9) FIG. **9** illustrates a cartridge according to an embodiment.
(10) FIG. **10** is a half exploded view illustrating the general configuration of an applicator according to an embodiment.
(11) FIG. **11** is a schematic diagram illustrating a drive mechanism according to an embodiment.
(12) FIG. **12** illustrates a configuration of the first drive mechanism according to an embodiment.
(13) FIG. **13** illustrates the second drive mechanism from one side according to an embodiment.
(14) FIG. **14** illustrates the second drive mechanism from the other side according to an embodiment.
(15) FIG. **15** illustrates an initial posture of a drive mechanism according to an embodiment.
(16) FIG. **16** illustrates a ready posture of a drive mechanism according to an embodiment.
(17) FIG. **17** illustrates a pressurizing posture of a drive mechanism according to an embodiment.
(18) FIG. **18** is a half exploded view illustrating a drive mechanism according to another embodiment.
(19) FIG. **19** illustrates a state in which a drive mechanism is connected according to another embodiment.
(20) FIG. **20**A and FIG. **20**B illustrate a drive mechanism according to yet another embodiment.
(21) FIG. **21**A and FIG. **21**B illustrate a wedge-shaped microneedle patch used in the drive mechanism according to yet another embodiment.
(22) FIG. **22** illustrates a control configuration of an applicator according to an embodiment.
(23) FIG. **23** is a flowchart illustrating a method of controlling an applicator performed by a controller according to an embodiment.
(24) FIG. **24** illustrates the first drug injection operation of an applicator according to an embodiment viewed from the back of the applicator.
(25) FIG. **25** illustrates the second drug injection operation of an applicator according to an embodiment viewed from the back of the applicator.
(26) FIG. **26** illustrates a touch sensor according to an embodiment.

(27) FIG. **27**A and FIG. **27**B are graphs illustrating an electrical signal measured by a touch sensor according to the mounting state of an applicator.

BEST MODE

(28) A microneedle applicator according to an embodiment includes: a patch delivery part configured to place a microneedle patch in a predetermined area; a housing having an cartridge accommodating part defined as a space for accommodating the cartridge and at least one opening; and a pressurizing part configured to apply a predetermined pressure to the microneedle patch, wherein the pressurizing part moves between a first position and a second position, and when the pressurizing part is at the second position, the pressurizing part comes into contact with the microneedle patch disposed in the predetermined area and applies the predetermined pressure to at least a portion of the microneedle patch, and the microneedle patch to which the predetermined pressure is applied is delivered directly to the skin through the opening.

MODE FOR INVENTION

(29) Hereinafter, specific exemplary embodiments of the present disclosure will be described in detail with reference to the accompanying drawings. However, the spirit of the present disclosure is not limited to the embodiments presented, and those skilled in the art who understand the spirit of the present disclosure may be able to easily propose other conventional inventions or other embodiments that fall within the scope of the spirit of the disclosure by adding, changing, or deleting other elements within the scope of the same spirit, but this will also be included within the scope of the disclosure.

(30) In addition, components that are the same in function within the scope of the same idea shown in the drawings of each embodiment will be described using the same reference numerals.

(31) A microneedle applicator according to an embodiment includes: a first drive mechanism configured to operate to pressurize a microneedle patch; a second drive mechanism operatively connected with the first drive mechanism to change a posture of the first drive mechanism; and a controller configured to transmit a driving signal to the first and second drive mechanisms, wherein the first drive mechanism includes: a first driving motor; and a pressurizing part configured to pressurize at least one region of the microneedle patch by receiving power from the first driving motor, the second drive mechanism includes: a second driving motor; and a main gear connected with the first drive mechanism to rotate by receiving power from the second driving motor, the controller actuates the second drive mechanism so that the pressurizing part moves from a first position to a second position of the microneedle patch and actuates the first drive mechanism so that the pressurizing part pressurizes the microneedle patch on the second position, and the pressurizing part may execute translation movement by 1 displacement substantially vertically downward on a surface of the microneedle patch according to an operation of the first driving motor.

(32) The second drive mechanism may be actuated for the pressurizing part to return from the second position to the first position.

(33) The controller may actuate the second drive mechanism so that the pressurizing part returned to the first position moves to a third position and actuate the first drive mechanism so that the pressurizing part pressurizes the microneedle patch on the third position.

(34) Also, the controller may actuate the second drive mechanism so that the pressurizing part returns from the third position to the first position.

(35) When the second drive mechanism is actuated for the pressurizing part to move from the first position to the second position, the main gear may rotate by second displacement along a first direction according to an operation of the second driving motor.

(36) When the second drive mechanism is actuated for the pressurizing part to return from the second position to the first position, the main gear may rotate by the second displacement along a second direction, which is different from the first direction, according to an operation of the second driving motor.

(37) Also, a direction of motion of the pressurizing part and a direction of rotation plane of the main gear may be vertical.
(38) The microneedle applicator further includes a touch sensor configured to be electrically connected to the controller and generate an electrical signal reflecting the state of contact with an object, wherein the controller may transmit a driving signal to the drive mechanism when a magnitude of the electrical signal generated by the touch sensor is greater than or equal to a threshold value.
(39) Also, the second drive mechanism includes a lower plate to protect the microneedle patch, and the lower plate is located on a lower part of the microneedle patch and may rotate with the main gear and the pressurizing part when the microneedle patch is mounted on the applicator.
(40) The lower plate includes a pressurizing-opening part, locations of the pressurizing-opening part and the pressurizing part correspond to each other, and when the first drive mechanism is actuated, at least a portion of the pressurized microneedle patch may protrude through the pressurizing-opening part.
(41) According to another embodiment of the present disclosure, a microneedle applicator includes: a patch delivery part configured to place a microneedle patch in a predetermined area; a housing including an accommodating part defined as a space for accommodating the patch delivery part and at least one opening; and a pressurizing part configured to apply a predetermined pressure to the microneedle patch, wherein the pressurizing part moves between a first position and a second position, and when the pressurizing part is at the second position, the pressurizing part comes into contact with the microneedle patch disposed in the predetermined area and applies the predetermined pressure to at least a portion of the microneedle patch, and the microneedle patch to which the predetermined pressure is applied may be delivered directly to the skin through the opening.
(42) The microneedle applicator may further include a strip for securing the applicator to a user's body.
(43) The patch delivery part includes a cartridge; the cartridge contains a plurality of microneedles; and the plurality of microneedles may be used a predetermined number of times for injections.
(44) Also, the microneedle applicator further includes a driving part, wherein the driving part may provide a driving force to at least one of the patch delivery part and the pressurizing part.
(45) The driving part may include at least one or more motor.
(46) Also, the microneedle patch has a preset dose of a drug stored at least a portion of therein; the applicator further comprises a controller; the controller may control at least one of the driving part, the patch delivery part, and the pressurizing part so that a preset dose stored in at least a portion of the microneedle patch may be injected into the user's skin.
(47) The controller may control the pressurizing part to move between the first position and the second position and to apply the preset pressure to the microneedle patch at the second position.
(48) At the second position, at least a portion of the microneedle patch pressurized by a predetermined displacement by the pressurizing part may be exposed through the opening.
(49) At the first position, the pressurizing part does not pressurize the microneedle patch and at least a portion of the microneedle patch may not be exposed.
(50) In addition, the applicator may be offered as a wearable watch.
(51) According to yet another embodiment of the present disclosure, a microneedle patch includes a body in which at least one opening is formed, and a microneedle body which is disposed in the opening and has at least one microneedle formed therein, wherein when an external force is applied in a first direction, the microneedle body moves in the first direction so as to administer a drug in the microneedle to a user's skin.
(52) The microneedle patch further include a connecting part, and when an external force is applied in a first direction on the microneedle body located at a first position, the microneedle body is moved to a second position by the elastic deformation of the connecting part, and the microneedle

body moved to the second position may move in a second direction opposite to the first direction.
(53) The microneedle body, which has moved to the second position, may move back to the first position in the second direction.
(54) Hereinafter, the general configuration of a drug delivery device according to an embodiment will be described with reference to FIGS. **1** and **2**.
(55) FIG. **1** illustrates the overall appearance of a drug delivery device according to an embodiment and FIG. **2** is a half exploded view illustrating the general configuration of a drug delivery device according to an embodiment.
(56) A description will be provided with reference to FIGS. **1** and **2**.
(57) According to an embodiment of the present disclosure, a drug delivery device **1** capable of supplying a drug to a user may be provided. That is, the drug delivery device **1** according to an embodiment may deliver the drug to the user according to the user's use.
(58) To this end, the drug delivery device according to an embodiment may be mounted on the user's body. In other words, the drug delivery device **1** in the present specification may be implemented in the form of a wearable device. To be provided in the form of a wearable device, all means for attaching the drug delivery device **1** to the user's body, such as bands and strips, may be provided together. The drug delivery device **1** may be mounted on the user's wrist, ankle, arm, leg, and etc., and it will be understood that it may be mounted on any body part suitable for supplying a drug to the user.
(59) Also, the drug delivery device **1** according to an embodiment may include an applicator **200** and a cartridge **100**. That is, in the present specification it was expressed as the drug delivery device **1** including the applicator **200** and the cartridge **1000**, but this is only for convenience of description and it should be understood that the applicator **200** and the drug delivery device **1** may be used interchangeably.
(60) The applicator **200** may accommodate the microneedle patch **100**. That is, a microneedle patch **100** may be mounted in the applicator **200**.
(61) When the microneedle patch **100** is mounted in the applicator **200**, the applicator **200** may deliver the drug stored in the microneedle patch **100** to the inside of the user's body.
(62) Hereinafter, with reference to the drawings, the microneedle patch according to the embodiment will be described first.
(63) FIG. **3** is a front stereoscopic view of a microneedle patch according to the first embodiment, FIG. **4** is a rear stereoscopic view of a microneedle patch according to the first embodiment, FIG. **5** is a cross-sectional view of a microneedle patch according to the first embodiment, and FIG. **6** is a bottom view of a microneedle patch according to the first embodiment.
(64) Referring to FIGS. **3** to **6**, the microneedle patch **100** according to the first embodiment may include a body **110**, an opening **120**, a shaft connecting part **130**, an auxiliary area **140**, and a microneedle structure body **150**.
(65) The microneedle patch **100** according to the first embodiment may be inserted into a watch-type wearable device. When inserted into a wearable device, the microneedle patch **100** may be operated in combination with the driving structure implemented in the wearable device. For example, the microneedle patch **100** may be rotated by the wearable device, and a portion may be moved in the first direction by being pressurized by the driving structure of the microneedle patch **100**.
(66) In addition, the microneedle patch **100** is fixed to the wearable device, and a portion of the micro patch **100** may be moved in the first direction when rotated and pressurized by the driving structure of the microneedle patch **100**.
(67) The microneedle patch **100** may have a replaceable structure. When the microneedle patch **100** has a replaceable structure, the microneedle patch **100** may be combined with other hardware structures to form a cartridge assembly. In this case, the microneedle patch **100** may be mounted on the cartridge assembly and inserted into the wearable device. The cartridge assembly may be

replaced when the injection of the drug attached to the microneedle patch **100** is completed or according to other needs.
(68) The body **110** may serve as a frame of the microneedle patch **100**. The body **110** may be a plate shape. The body **110** may be a circular plate shape. That is, the body **110** may be a thin disk shape.
(69) A plurality of openings **120** and a shaft connecting part **130** may be formed in the body **110**. The plurality of openings **120** may be formed to pass through the body **110**. The plurality of openings **120** may be arranged in a sectoral shape with respect to the center of the body **110**. The plurality of openings **120** may be located at a predetermined distance from the adjacent openings **120**. Alternatively, although not shown, the plurality of openings **120** may be arranged in various shapes such as a circle.
(70) The opening **120** may have a different area size depending on a distance from the center of the body **110**. For example, an area of a region adjacent to the center of the body **110** among the openings **120** may be smaller than an area of a region spaced apart from the center **110** of the body. The opening **120** may be formed to have a larger area as the distance from the center of the body **110** increases.
(71) The shaft connecting part **130** may be formed to pass through the body **110**. The shaft connecting part **130** may be formed in the center of the body **110**. The shaft connecting part **130** may be formed in a circular shape based on the center of the body **110**. That is, the center of the shaft connecting part **130** may be the same as that of the body **110**. The outer diameter of the shaft connecting part **130** may be smaller than that of the body **110**. The plurality of openings **120** may be formed between the outer diameter of the shaft connecting part **130** and the outer diameter of the body **110**.
(72) An external structure for fixing or rotating the microneedle patch **100** may be inserted into the shaft connecting part **130**. The external structure may be a partial structure of the wearable device, or may be a partial structure of the cartridge assembly. The external structure may rotate the microneedle patch **100** by an operating structure of the wearable device. On the other hand, the microneedle patch **100** may be fixed to the wearable device by a structure coupled to the body **110**.
(73) An auxiliary region **140** may be formed in the body **110**. The auxiliary region **140** may be a region in which the opening **120** is not formed. The auxiliary region **140** may be an unopened region of the body **110**. The plurality of openings **120** are located at regular intervals, and the opening **120** may not be formed in an area where one opening **120** can be located, and an area in which the opening **120** is not formed may be defined as the auxiliary region **140**. The auxiliary region **140** may serve as a door blocking the cartridge assembly from the outside when the wearable device is in a standby state rather than a drug injection state. The auxiliary region **140** may serve to block the internal structure of the wearable device from the outside in the standby state. By the auxiliary region **140**, the cartridge assembly and the internal structure of the wearable device may be protected from external moisture or foreign substances.
(74) The microneedle structure body **150** may be located in the opening **120**. The microneedle structure body **150** may be moved when an external force is applied. The microneedle structure body **150** may move in the same direction with the external force when the external force is applied, then return to the original position again. The microneedle structure body **150** may move in the same direction with the external force when the external force is applied, and when the external force is released, may return to the original position again.
(75) For example, when the external force is applied in the first direction while the microneedle structure body **150** is positioned at the first position, the microneedle structure body **150** may move in the first direction to the second position, then return to the first position by moving in the second direction opposite to the first direction. The microneedle structure body **150** moved to the second position may move in the second direction when the external force is released, and return to the first position. The first position may be an initial position of the microneedle structure body **150**.

(76) Since the microneedle structure body **150** has an elastic structure, it can move in the first direction by the external force, and can move again in the second direction when the external force is released.
(77) The external force may be applied to the microneedle structure body **150** by the operating structure of the wearable device, or the external force may be applied to the microneedle structure body **150** by the operating structure of the cartridge assembly. The external force may be transmitted to the microneedle structure body **150** by a power source located inside the wearable device.
(78) The microneedle structure body **150** may include a microneedle body **151** and a connecting part **155**.
(79) The microneedle body **151** may be formed in a plate shape having both sides.
(80) The microneedle body **151** may include a plurality of microneedles **153**. The plurality of microneedles **153** may be formed on one surface of the microneedle body **151**. The plurality of microneedles **153** may be formed to protrude from the microneedle body **150** in the first direction.
(81) The plurality of microneedles **153** may serve to deliver drugs to the user's skin. The plurality of microneedles **153** may be composed of at least one type of a coated type, a hollow type, a dissolving type, and a swelling type.
(82) When the microneedle **153** is a coated type, the outside of the microneedle **153** is coated with the drug and when the microneedle is injected into the user's skin, the drug coated on the microneedle **153** can be delivered to the user's skin.
(83) When the microneedle **153** is a hollow type, a hollow is formed in the microneedle **153** and when the microneedle is injected into the user's skin, the drug that was injected in the hollow can be delivered to the user's skin.
(84) When the microneedle **153** is a dissolving type, the microneedle **153** itself is formed of a drug and when the microneedle is injected into the user's skin, the microneedle **153** itself is delivered to the user, and then the microneedle is dissolved in the user's skin so that the drug can be delivered to the user's skin.
(85) When the microneedle **153** is a swelling type, a swellable material containing a drug is coated on the outside of the microneedle **153**, and when the microneedle is injected into the user's skin, the coated material swells so that the drug can be structurally bonded to the user's skin and delivered to the user.
(86) An external force may be applied to the other surface of the microneedle body **151**. When the external force is applied to the other surface of the microneedle body **151** in the first direction, a plurality of microneedles **153** formed on one surface of the microneedle body **151** may move in the direction of the user's skin to deliver the drug to the user's skin.
(87) The connecting part **155** may connect the microneedle body **151** and the body **110**. The connecting part **155** may be formed in a structure having elasticity. The connecting part **155** may be formed in a structure that can be elastically deformed. However, the elastic deformation may be a temporary deformation.
(88) When an external force is applied to the microneedle body **151** in the first direction, the connecting part **155** may be elastically deformed so that the microneedle body **151** moves in the first direction while the microneedle body **151** maintains the connection with the body **110**. Also, when the external force is released, the microneedle body **151** may be moved in the second direction by the elasticity of the connecting part **155**.
(89) The plurality of microneedle bodies **151** included in the microneedle patch **100** are rotated by the elasticity of the connecting part **155** to deliver the drug to the skin once per rotation. As such, the microneedle patch **100** has the effect of performing drug delivery multiple times.
(90) The connecting part **155** may include a first connecting part **157** and a second connecting part **159**.
(91) The first connecting part **157** may be located in an area adjacent to the shaft connecting part

130. The second connecting part **159** may be located in an area spaced apart from the shaft connecting part **130**. The first connecting part **157** may be located in a region adjacent to the center of the body **110**, and the second connecting part **159** may be located in an area adjacent to the outer diameter of the body **110**.
(92) The first connecting part **157** may connect one side of the microneedle body **151** and the body **110**, and the second connecting part **159** may connect the other side of the microneedle body **151** to the body **110**.
(93) The first connecting part **157** and the second connecting part **159** may be formed in a structure having a curved shape. The first connecting part **157** and the second connecting part **159** may have elasticity by being formed in a structure having a curved shape.
(94) The first connecting part **157** and the second connecting part **159** may be designed as a structure having different elastic forces. The first connecting part **157** and the second connecting part **159** may be formed in a structure having elasticity to move in the first direction when an external force is applied to the microneedle body **151** in the first direction. That is, the first connecting part **157** and the second connecting part **159** may be designed in a structure having elasticity so that it can move to the second position in a state parallel to the upper surface of the body **110** when an external force is applied in the first direction.
(95) The first connecting part **157** and the second connecting part **159** may be formed in a spring structure.
(96) The first connecting part **157** and the second connecting part **159** each may be formed in a curved structure having a width and a pitch. The first connecting part **157** may have a structure having a width and a pitch different from that of the second connection part **159**.
(97) The first connecting part **157** may have a first width w**1** and a first pitch p**1**. The first width w**1** may be defined as a distance between the regions of the first connecting part **157** that is farthest apart based on an imaginary line connecting the center of the microneedle body **151** to the center of the body **110**. The first pitch p**1** may be defined as an distance between adjacent regions within the first connection part **157**. That is, the first pitch p**1** may be defined as a minimum distance between adjacent regions within the first connecting part **157** based on an imaginary line connecting the center of the microneedle body **151** and the center of the body **110**.
(98) The second connecting part **159** may have a second width w**2** and a second pitch p**2**. The second width w**2** and the second pitch p**2** may also be defined as a meaning corresponding to the first width w**1** and the first pitch w**2** of the first connection part **157**.
(99) The first width w**1** may be smaller than the second width w**2**. The first pitch p**1** may be greater than or equal to the second pitch p**2**.
(100) In addition, the first connecting part **157** and the second connecting part **159** may have a curved section and a straight section. In this case, the width of the first connecting part **157** and the second connecting part **159** may be defined as an interval between a straight line connecting inflection points of one curved section and a straight line connecting inflection points of the other curved section. The pitch of the first connecting part **157** and the second connecting part **159** may be defined as an interval between the straight sections.
(101) The microneedle body **151**, the first connecting part **157**, and the second connecting part **159** may be integrally formed. The microneedle body **151**, the first connecting part **157**, and the second connecting part **159** may have the same thickness. Alternatively, although not shown, the thickness of the microneedle body **151** may be greater or smaller than the thickness of the first connecting part **157** and the second connecting part **159**.
(102) The body **110** and the microneedle structure body **150** may be integrally formed. Also, the body **110**, the microneedle structure body **150**, and the microneedle **153** may be integrally formed.
(103) A protective film **160** may be formed on at least one surface of the body **110**. The protective film **160** may serve to protect the microneedle body **151** from the outside.
(104) The protective film **160** may include an upper protective film **161** and a lower protective film

163. The upper protective film **161** may be formed on the upper surface of the body **110**, and the lower protective film **163** may be formed on the lower surface of the body **110**.
(105) The protective film **160** may be made of a material including metal. The protective film **160** may be made of aluminum. Also, the protective film **160** may be made of plastics.
(106) The protective film **160** may be damaged by the microneedle **151** when the microneedle structure body **150** is pressurized by the external structure, so that the microneedle **151** may deliver a drug to the user's skin.
(107) Alternatively, when drug injection from a specific microneedle structure body **150** among a plurality of microneedle structure bodies **150** is scheduled, the protective film **160** in the area corresponding to the specific microneedle structure **150** may be damaged or removed in advance just before pressurization by the external structure, so that drug injection by the microneedle **151** can be performed more smoothly. In this case, a portion of the protective film **160** may be damaged, moved, or removed by some structures of the cartridge assembly or the wearable device.
(108) FIG. **7** is a bottom perspective view illustrating a patch assembly according to the second embodiment.
(109) The patch assembly according to a second embodiment is provided in a form in which a microneedle patch is mounted on a patch carrier without a separate cartridge housing. Therefore, in the description of the patch assembly according to a second embodiment, the same reference numerals are assigned to the descriptions common to those of a first embodiment, and detailed descriptions thereof are omitted.
(110) Referring to FIG. **7**, the patch assembly according to the second embodiment includes a microneedle patch **100** and a patch carrier **200**.
(111) The microneedle patch **100** may be mounted on the patch carrier **200**. The microneedle patch **100** may be rotatably mounted on the patch carrier **200**.
(112) The microneedle patch **100** may include a body **110**, a plurality of openings **120**, a shaft connecting part **130**, an auxiliary region **140**, and a microneedle structure body **150**.
(113) The patch carrier **200** may include a carrier body **210** and a coupling structure **220**.
(114) The carrier body **210** constitutes a frame of the patch carrier **200**. A seating part may be formed on the carrier body **210**, and the microneedle patch **100** may be rotatably mounted to the seating part.
(115) The coupling structure **220** may be formed in a portion of the carrier body **210**. The coupling structure **220** may have a shape protruding to the outside. The coupling structure **220** may serve to fix the patch carrier **200** to the wearable device when the patch carrier **200** is mounted on the wearable device.
(116) A through groove **230** may be formed in the patch assembly. The through groove **230** may serve as a path through which a power transmission structure among the internal structures of the wearable device moves from the outside of the patch assembly to the shaft connecting part **130**. That is, when the patch assembly is inserted into the wearable device, the through groove **230** may be a moving path for the power transmission structure, which should be located in the shaft connecting part **130** when the installation is complete, to move to the shaft connecting part **130** without structural interference.
(117) The through groove **230** may include a patch through groove **170** and a carrier through groove **240**. The patch through groove **170** may be formed to extend from the shaft connecting part **130** to the outer diameter of the body **110**. The patch through groove **170** may be formed in the auxiliary region **140**.
(118) The carrier through groove **240** may be formed to pass through the seating part and the carrier body **210**. The patch through groove **170** and the carrier through groove **240** may be formed in corresponding regions.
(119) The patch assembly according to the second embodiment is provided in a form in which a microneedle patch is mounted on a patch carrier without a separate cartridge housing, so that it can

be implemented with a simple structure, thereby reducing the manufacturing cost. In addition, since the patch assembly has the through groove **230**, the power transmission structure of the wearable device can be positioned with the shaft connecting part **130** without structural interference, thereby omitting a separate movement mechanism.
(120) FIG. **8** is a top view illustrating a microneedle patch according to the third embodiment.
(121) The microneedle patch according to the third embodiment is different from the first embodiment in the shape of the connecting part, the microneedle body, and the shape of the body corresponding thereto, and the rest of the configuration is the same. Therefore, in the description of the third embodiment, the same reference numerals are assigned to the components common to the first embodiment, and detailed descriptions thereof are omitted.
(122) Referring to FIG. **8**, the microneedle patch **300** according to the third embodiment may include a body **310**, a shaft connecting part **330**, and a microneedle structure body **350**.
(123) The body **310** may include a first body **311** and a second body **313**. The first body **311** and the second body **313** may have a circular band shape. The first body **311** and the second body **313** may have a circular band shape having the same center. The first body **311** and the second body **313** may have different radii. The first body **311** may have a smaller radius than that of the second body **313**. The second body **313** may be formed to surround the first body **311**. The first body **311** and the second body **313** may be formed to be spaced apart from each other. The first body **311** and the second body **313** may be spaced apart from each other to form an opening.
(124) The microneedle structure body **350** may be located in an opening between the first body **311** and the second body **313**. At least one microneedle structure **350** may be positioned in the opening.
(125) The microneedle structure **350** may be connected to the first body **311** and the second body **313**.
(126) The microneedle structure **350** may include a microneedle body **351**, a connecting part **355**, and an annular connecting part **356**.
(127) A microneedle may be formed in the microneedle body **351**.
(128) The microneedle body **351** may be formed in a sectoral shape. In the drawings, the microneedle body **351** has been described as having a sectoral shape, but is not limited thereto.
(129) The widths of one end and the other end of the microneedle body **351** may be different. One end of the microneedle body **351** may be adjacent to the first body **311**, and the other end of the microneedle body **351** may be adjacent to the second body **313**. The width of one end of the microneedle body **351** may be smaller than the width of the other end.
(130) The microneedle body **351** may be connected to the main body **310** by the connecting part **355**. The connecting part **355** may be arranged toward the center of the circle.
(131) The connecting part **355** may include a first connecting part **357** and a second connecting part **359**.
(132) The first connecting part **357** may be positioned between the microneedle body **351** and the first body **311**. The first connecting part **357** may connect the first body **311** and the microneedle body **351**. The first connecting part **357** may have a first width W**1** and a first thickness T**1**.
(133) The second connecting part **359** may be positioned between the microneedle body **351** and the second body **313**. The second connecting part **359** may connect the second body **313** and the microneedle body **351**. The second connecting part **359** may have a second width W**2** and a second thickness T**2**.
(134) The first width W**1** may be different from the second width W**2**, and the first thickness T**1** may be different from the second thickness T**2**.
(135) The annular connecting part **356** may connect the adjacent microneedle body **351**. The annular connecting part **356** may connect the side surfaces of the adjacent microneedle body **351**. The annular connecting part **356** may be formed in a shape corresponding to the main body **310**. A plurality of annular connecting parts **356** may be formed to be spaced apart from each other along a circular band shape. The annular connecting part **356** may be disposed along a circle having the

same center as the first body **311**.
(136) The connecting part **355** and the annular connecting part **356** may be physically connected in a state formed separately from the microneedle body **351**, and at least one of the connecting part **355** and the annular connecting part **356** may be integrally formed with the microneedle body **351**.
(137) The connecting part **355** and the annular connecting part **356** may have elasticity. The connecting part **355** may have elasticity in the center direction of the body **310**. The annular connecting part **356** may have the same center as the body **310**, and may have elasticity in the circumferential direction of a circle in which the annular connecting part **356** is located.
(138) Because the connecting part **355** and the annular connecting part **356** have elasticity, when an external force is applied to the microneedle body **351** in the first direction, the microneedle body **351** may be moved to the second position in a state parallel to the upper surface of the body **310**. After moving to the second position, the microneedle body **351** may be moved back to the first position.
(139) FIG. **9** illustrates a cartridge according to an embodiment.
(140) Referring to FIG. **9**, the microneedle patch **100** may be provided through the cartridge **1000**. That is, the microneedle patch **100** may be mounted on the applicator **200** (see FIG. **2**) while being mounted or accommodated in the cartridge **1000**.
(141) According to an embodiment, the cartridge **1000** may be manufactured together with the microneedle patch **100** and used once. According to another embodiment, the cartridge **1000** may be implemented as a part of the applicator **200** (see FIG. **10**) to enable reuse.
(142) In the following description, the cartridge **1000** will be mainly described when implemented as a part of the applicator **200** (see FIG. **10**), but it should be noted in advance that this is only for convenience of description.
(143) The cartridge **1000** according to an embodiment may include a body portion **1002** and a flat plate portion **1004**. Here, the body portion **1002** may be accommodated in an accommodating part formed in the housing **201** (see FIG. **10**) of the applicator **200** (see FIG. **10**), as will be described later.
(144) In addition, the flat plate portion **1004** may have a flat plate opening **1006** through which the pressurizing part or the rack gear can pass, as will be described later.
(145) Also, the flat plate portion **1004** may be mounted on the body portion **1002**. The flat plate portion **1004** may be inserted into the applicator **200** (see FIG. **10**) while being mounted on the body portion **1002**. When the flat plate portion **1004** is inserted into the applicator **200** (see FIG. **10**), a fixing portion **1005** formed on the flat plate portion **1004** may be operatively coupled to the drive mechanism **210** (see FIG. **10**), as will be described later. When the fixing portion **1005** is operatively coupled to the drive mechanism **210** (see FIG. **10**), the flat plate portion **1002** may rotate according to the driving of the drive mechanism **210** (see FIG. **10**).
(146) In addition, the cartridge **1000** performs a function of placing the microneedle patch **100** (see FIG. **3**) in a predetermined area. The microneedle patch **100** (see FIG. **3**), generally a microneedle **153** (see FIGS. **3** to **7**), is disposed on each microneedle body **151** (see FIG. **3**), and since the microneedle included in each microneedle body stores a preset dose of a drug, it is important that the microneedle patch be placed at a predetermined location inside the applicator when injecting into the user through the applicator. To describe this from another point of view, the cartridge **1000** may be expressed as performing a function of delivering a patch to an appropriate location, and the cartridge **1000** may be expressed as a patch delivery unit.
(147) When the cartridge **1000** places the microneedle patch **100** at a predetermined location inside the applicator **200**, the applicator **200** may inject a predetermined dose to the user through the drive mechanism **210**.
(148) Interaction between the cartridge **1000** and the applicator **200** will be described later in detail.
(149) In addition, since an upper protective film **100***a* and a lower protective film **100***b* shown in the drawings are the same as the above-described upper protective film **161** (see FIG. **5**) and the

lower protective film **163** (see FIG. **5**), a detailed description thereof will be omitted.
(150) In the above, the micro-needle patch and the cartridge accommodating the micro-needle patch according to various embodiments were described. Hereinafter, an applicator for applying a microneedle patch according to an embodiment will be described.
(151) FIG. **10** is a half exploded view illustrating the general configuration of an applicator according to an embodiment.
(152) Referring to FIG. **10**, the applicator **200** according to an embodiment may include a housing **201**, a drive mechanism (or drive unit) **210**, a PCB **203**, and a cartridge **205**.
(153) First, the housing **201** serves to protect the components inside the applicator **200**, and for this purpose, it is formed to surround the outside of the applicator **200**. The shape of the housing **201** may vary according to a design change.
(154) The housing **201** may include an accommodating part **206** for accommodating the cartridge **205** as described above. The accommodating part **206** may be formed to correspond to the shape of the cartridge **205**. The accommodating part **206** may be formed in the form of an opening in the housing **201**, and an accommodating space corresponding to the shape of the cartridge **205** may be formed inside the housing **201**.
(155) Also, according to an embodiment, the housing **201** may have an open bottom shape, and in this case, the housing **201** may include a lower cover **202**. The lower cover **202** may be mounted on the open bottom of the housing **201**. The lower cover **202** may function to support the cartridge **205**. As will be described later, the lower cover **202** may include an opening so that, when the microneedle patch **100** (see FIG. **3**) is pressurized according to the operation of the drive mechanism **210**, the microneedle patch **100** (see FIG. **3**) comes in contact with the user's body.
(156) The drive mechanism **210** may be mounted inside the housing **201**. The drive mechanism **210** may include at least one or more motors and gears. The drive mechanism **210** provides power for the applicator **200** to pressurize the microneedle patch **100** (see FIG. **3**). A detailed description of the drive mechanism **210** will be described in detail with reference to FIGS. **11** to **17**.
(157) The PCB **203** is electrically connected to the drive mechanism **210** and may be mounted inside the housing **201**. The PCB **203** may include an MCU, and the drive mechanism **210** may operate as a driving signal generated by the MCU is transmitted to the drive mechanism **210**. Also, the overall operation of the applicator **200** can be controlled according to the design of an electrical circuit formed on the PCB.
(158) In addition, the applicator **200** includes a power source for receiving power. The power source may be electrically connected to the PCB **203**. The powered PCB may generate a driving signal and transmit it to the drive mechanism **210**, and the drive mechanism **210** that receives the driving signal operates so that the applicator **200** pressurizes the microneedle patch **100** (see FIG. **3**) to provide a drug to the user.
(159) The applicator **200** according to an embodiment may include a display module **204**. Various indications may be provided to the user through the display module **204**.
(160) Hereinafter, the operation of the drive mechanism **210** will be described with reference to the drawings.
(161) FIG. **11** is a schematic diagram illustrating a drive mechanism according to an embodiment.
(162) According to an embodiment, the drive mechanism **210** may pressurize at least a portion of the microneedle patch **100** (see FIG. **3**).
(163) Referring to FIG. **11**, the drive mechanism **210** may include a first drive mechanism **400** and a second drive mechanism **500**. Here, the first drive mechanism **400** and the second drive mechanism **500** may be operatively connected through the connecting unit **440**. Also, the first drive mechanism **400** and the second drive mechanism **500** may be operatively connected through a shaft **450** (see FIG. **12**)
(164) First, the first drive mechanism **400** according to an embodiment may perform an operation of pressurizing at least a portion of the microneedle patch **100**. The second drive mechanism **500**

may perform an operation of changing the posture or position of the first drive mechanism **400** so that the first drive mechanism **400** can pressurize an appropriate point of the microneedle patch **100**.
(165) Specifically, when the second drive mechanism **500** operates, the first drive mechanism **400** may move in response to the operation of the second drive mechanism **500**. When the first drive mechanism **400** has a posture for pressurizing the microneedle patch **100** by moving in response to the operation of the second drive mechanism **500**, the first drive mechanism **400** is operated to pressurize the microneedle patch **100**.
(166) Hereinafter, a specific configuration of the first drive mechanism **400** will be described with reference to the drawings.
(167) FIG. **12** illustrates a configuration of the first drive mechanism according to an embodiment.
(168) Referring to FIG. **12**, the first drive mechanism **400** according to an embodiment may include a first driving motor **402**, at least one first connecting gear **403**, and a pressurizing part **404**.
(169) The first driving motor **402** may be engaged with at least one or more first connecting gears **403**, and at least one or more of the first connecting gears **403** may be operatively connected to the pressurizing part **404**. Here, the gear structure of the at least one or more first connecting gears **403** may be variously changed in design to achieve the object of the present disclosure, and it will be apparent that such modifications are also incorporated within the spirit of the present disclosure.
(170) The first driving motor **402** may operate by receiving a driving signal and power from the above-described PCB. When the first driving motor **402** operates, at least one or more first connecting gears **403** may be engaged with the first driving motor **402** to operate. When at least one first connecting gear **403** that has received the power of the first driving motor **402** operates, the pressurizing part **404** operatively connected with the first connecting gear **403** may operate.
(171) The pressurizing part **404** may pressurize at least a portion of the microneedle patch **100** by moving translationally along an axis perpendicular to the surface of the microneedle patch **100**. That is, by operating the first drive mechanism **400** according to the driving signal received from the PCB, the pressurizing part **404** can pressurize at least a portion of the microneedle patch **100**. Here, the pressurizing part **404** may move by a predetermined displacement according to the operation of the first drive mechanism **400**. In other words, it may be expressed that the pressurizing part **404** can move from the first position to the second position according to the operation of the first drive mechanism **400**. Also, the pressurizing part **404** applies a predetermined pressure to at least a partial region (e.g., the microneedle body) of the microneedle patch **100** (see FIG. **3**) by moving a predetermined displacement or moving between predetermined positions.
(172) In addition, the first drive mechanism **400** may be connected to the second drive mechanism **500** through the shaft **450** and the connecting part **440**. As will be described later, the shaft **450**, which is the axis of rotation of the second drive mechanism **500**, and the first drive mechanism **400** are connected, so that the first drive mechanism **400** rotates as a whole relative to the shaft **450** according to the operation of the second drive mechanism **500**, and the pressurizing part **404** may pressurize the microneedle patch **100** at a suitable location.
(173) Hereinafter, an operation of the second drive mechanism **500** will be described with reference to the drawings.
(174) FIGS. **13** to **14** illustrate a second drive mechanism **500** according to an embodiment.
(175) FIG. **13** illustrates the second drive mechanism from one side according to an embodiment, and FIG. **14** illustrates the second drive mechanism from the other side according to an embodiment.
(176) Referring to FIGS. **13** to **14**, the second drive mechanism **500** according to an embodiment may include a second driving motor **502**, a main gear **504**, one or more second connecting gears **505** and **506**, and a lower plate **510**.
(177) First, the second driving motor **502** may operate by receiving a driving signal and power from the above-described PCB **203** (see FIG. **10**). When the second driving motor **502** operates, the

power of the second driving motor **502** may be transmitted to one or more second connecting gears **505** and **506** as a transmission gear **508** connected to the second driving motor **502** operates. One or more second connecting gears **505** and **506** that have received the power of the second driving motor **502** may engage and operate with the main gear **504**. That is, one or more second connecting gears **505** and **506** transmit the power of the second driving motor **502** to the main gear **504**, so that the main gear **504** may rotate around the shaft **450**. Here, one or more second connecting gears **505** and **506** may perform a function of controlling the rotational speed of the main gear **504** according to a setting. It will be understood by those skilled in the art that in order to control the rotational speed of the main gear **504** according to the output of the second driving motor **502** or the driving signal, one or more second connecting gears **505** and **506** may have various design modifications of the gear structure, and it will be appreciated that such design variations are also incorporated within the spirit of this specification.
(178) The main gear **504** may be coupled to the shaft **450**. The shaft **450** may be coupled to the lower plate **510**. That is, the main gear **504** and the lower plate **510** are coupled to both sides of the shaft **450** respectively so that the main gear **504** and the lower plate **510** may rotate to correspond to each other.
(179) Here, the lower plate **510** may include a pressurizing opening part **512**. The pressurizing part **404** may pressurize the microneedle patch **100** (see FIG. **3**) through the pressurizing opening part **512**. That is, as will be described later, when the microneedle patch **100** (see FIG. **3**) is inserted into the applicator **200** (see FIGS. **3** to **10**), the microneedle patch **100** may be positioned on the lower plate **510**. At this time, when the pressurizing part **404** pressurizes the microneedle patch **100** (see FIG. **3**), the microneedle may contact the user's body through the pressurizing opening part **512**.
(180) The first drive mechanism **400** and the second drive mechanism **500** may be operatively connected to each other through a mechanism connecting part **440**.
(181) Specifically, the first drive mechanism **400** may include the mechanism connecting part **440**. The mechanism connecting part **440** may be coupled to the mechanism accommodating part **540** formed in the main gear **504**. The mechanism connecting part **440** may also be connected to the shaft **450**. The mechanism connecting part **440** may further include a shaft accommodating part (not shown), and the shaft **450** may be coupled to the shaft accommodating part (not shown).
(182) As the mechanism connecting part **440** is connected to the mechanism accommodating part **540** and the shaft **450**, when the main gear **504** rotates, the first drive mechanism **400** may rotate as a whole in response to the rotation of the main gear **504**.
(183) As such, the lower plate **510** and the first drive mechanism **400** may rotate to correspond to the main gear **504** according to the operation of the second driving motor **502**.
(184) Here, the applicator **200** (see FIG. **10**) according to an embodiment may be set such that the initial position of the pressurizing part **404** of the first drive mechanism **400** corresponds to the pressurizing opening part **512** of the lower plate **510**. For this reason, even when the main gear **504** rotates due to the operation of the second drive mechanism **500**, the positions of the pressurizing part **404** and the pressurizing opening part **512** may correspond to each other. Thereafter, when the first drive mechanism **400** operates, the pressurizing part **404** pressurizes at least a portion of the microneedle patch **100** so that the microneedle can contact the user's body through the pressurizing opening part **512** and the opening of the lower cover **202**.
(185) FIGS. **15** to **17** illustrate the operation of a drive mechanism according to an embodiment.
(186) FIG. **15** illustrates an initial posture of a drive mechanism according to an embodiment, FIG. **16** illustrates a ready posture of a drive mechanism according to an embodiment, and FIG. **17** illustrates a pressurizing posture of a drive mechanism according to an embodiment.
(187) The drive mechanism **210** according to an embodiment may change the posture to pressurize the microneedle patch **100**. Specifically, the second drive mechanism **500** may change the position of the first drive mechanism **400** such that the first drive mechanism **400** can pressurize the microneedle patch **100** at a suitable position.

(188) It will be described as a specific example with reference to the drawings.
(189) Referring to FIG. **15**, in the initial state (or standby state) of the applicator **200**, the pressurizing part **404** may be located in a position corresponding to the auxiliary region **140** (see FIG. **4**) of the microneedle patch **100**. At this time, the pressurizing opening part **512** of the lower plate **510** may also be located at a position corresponding to the auxiliary region **140** (see FIG. **4**).
(190) In other words, it can be expressed that the pressurizing part **404** is located above the auxiliary region **140** (see FIG. **4**), and the pressurizing opening part **512** may be located below the auxiliary region **140** (see FIG. **4**).
(191) That is, when the applicator **200** is in the standby state, if the position corresponding to the auxiliary region **140** (see FIG. **4**) is expressed as the first position, both the pressurizing part **404** and the pressurizing opening part **512** may be located in the area corresponding to the first position.
(192) As mentioned above, the auxiliary region **140** (see FIG. **4**) prevents foreign substances from entering the microneedle patch **100** (see FIGS. **3** to **7**) or the housing **201** (see FIG. **10**) when the applicator **200** (see FIG. **10**) is in a standby state, and correspondingly, the pressurizing part **404** and the pressurizing opening part **512** may be located at positions corresponding to the auxiliary region **140** (see FIG. **4**) when the applicator **200** (see FIG. **10**) is in a standby state.
(193) Referring to FIGS. **16** to **17**, the state in which the applicator **200** (see FIG. **10**) has entered into an operating state is shown. When the applicator **200** (see FIG. **10**) enters the operating state, the main gear **504** rotates according to the operation of the second drive mechanism **500**. When the main gear **504** rotates, the position of the pressurizing part **404** rotating corresponding to the main gear **504** is also changed. The pressurizing part **404** rotates around the shaft **450** (see FIG. **14**) according to the rotation of the main gear **504** to be located on a position corresponding to the microneedle body **151** (see FIG. **3**). In this case, in response to the rotation of the main gear **504**, the lower plate **510** also rotates, so that the pressurizing opening part **512** is also located at the bottom of the microneedle body **151** (see FIG. **3**).
(194) That is, the microneedle patch **100** (see FIG. **3**) does not rotate, whereas the pressurizing part **404** and the pressurizing opening part **512** rotate with the microneedle patch **100** (see FIG. **3**) interposed therebetween so that the pressurizing part **404** and the pressurizing opening part **512** will be placed in position corresponding to the microneedle body **151**.
(195) To put it another way, if the position corresponding to the microneedle body **151** (see FIG. **3**) is expressed as the second position, it may also be said that when the applicator **200** (see FIG. **10**) is in an operating state, the pressurizing part **404** and the pressurizing opening part **512** may also be placed in a region corresponding to the second position.
(196) By placing the pressurizing part **404** and the pressurizing opening part **512** at positions corresponding to the microneedle body **151** (see FIG. **3**), the microneedle is ready to contact the user's skin.
(197) Referring to FIG. **17**, the pressurizing part **404** executes translation movement vertically downward to the microneedle patch **100** (see FIG. **3**) or the lower plate **510** according to the operation of the first drive mechanism **400**. Here, as described above, the pressurizing part **404** may apply a predetermined pressure to the microneedle patch by moving a predetermined displacement. As a result, the microneedle body **151** (see FIG. **3**) is pressurized, and the microneedle located at the lower part of the pressurized microneedle body **151** (see FIG. **3**) may protrude to the outside of the pressurizing opening part **512**. Accordingly, the microneedle located at the lower part of the microneedle body **151** (see FIG. **3**) may be injected into the user's skin.
(198) When the pressurizing part **404** returns to the initial position according to the driving of the first drive mechanism **400** after the microneedle is injected into the user's skin, the second drive mechanism **500** operates to return the pressurizing part **404** and the pressurizing opening part **512** present in the second position to the first position. Specifically, the main gear **504** rotates according to the operation of the second driving motor **502**, and the pressurizing part **404** and the pressurizing opening part **512** rotate around the shaft **450** in response to the rotation of the main gear **504**,

thereby returning to the first position.
(199) When the pressurizing part **404** and the pressurizing opening part **512** are returned to the first position, the applicator **200** enters the standby state. As will be described later, when the applicator **200** enters the operating state again, the second drive mechanism **500** operates to position the pressurizing part **404** and the pressurizing opening part **512** on the microneedle body corresponding to the third position. Here, the third position will mean a position different from the microneedle body corresponding to the second position. Also, the microneedle body corresponding to the second position and the microneedle body corresponding to the third position may be adjacent to each other. Also, a distance from the first location to the third location may be longer than a distance from the first location to the second location.
(200) FIGS. **18** and **19** illustrate a drive mechanism according to another embodiment.
(201) FIG. **18** is a half exploded view illustrating a drive mechanism according to another embodiment, and FIG. **19** illustrates a state in which a drive mechanism is connected according to another embodiment.
(202) Referring to FIG. **18**, the applicator **200** according to an embodiment may include a third drive mechanism **600** and a fourth drive mechanism **700**.
(203) The third drive mechanism **600** is functionally similar to the first drive mechanism **400** (see FIGS. **13** to **17**), but there is a difference in the mechanism structure thereof. In the description of this specification, only the differences between the third drive mechanism and the first drive mechanism **400** (see FIGS. **13** to **17**) will be described, and it will be understood that other descriptions may refer to the contents regarding the first driving mechanism **400**.
(204) The fourth drive mechanism **700** is functionally similar to the second drive mechanism **500** (see FIGS. **13** to **17**), and there is a difference in the mechanism structure thereof. Again, differences between the second drive mechanism and the fourth drive mechanism **700** will be mainly described.
(205) The third drive mechanism **600** according to an embodiment may include a third driving motor **602** and a transmission gear (not shown) and a rack gear **604**.
(206) Since the description of the third driving motor **602** is similar to the above-description of the first driving motor **402**, a detailed description thereof will be omitted.
(207) Unlike the structural relationship between the first drive mechanism **400** and the second drive mechanism **500**, the third drive mechanism **600** may be mainly located on top of the fourth drive mechanism **700**. Specifically, the third driving motor **602** may be disposed on top of the frame gear **706**. The rack gear **604** may move in a downward direction of the applicator **200** during operation through an opening formed in the frame gear **706**. Specifically, the rack gear **604** may be mounted on a rack gear accommodating part **707** formed on the frame gear **706**.
(208) When the third driving motor **602** operates, one or more transmission gears (not shown) may transmit the power of the third driving motor **602** to the rack gear **604**. The rack gear **604** receiving power from the third driving motor **602** executes vertical translation movement in the rack gear accommodating part **707**, thereby pressurizing the microneedle body **151** (see FIG. **3**). Here, a separate pressurizing region may be set in the rack gear **604**, or an additional pressurizing portion may be added.
(209) The third driving motor **602** and the one or more transmission gears may be protected through a protection part **606**.
(210) The fourth drive mechanism **700** includes a fourth driving motor **702**, an auxiliary gear **704** and a frame gear **706**.
(211) The description of the fourth driving motor **702** is similar to the description of the second driving motor **502** and thus will be omitted.
(212) The auxiliary gear **704** receives power from the fourth driving motor **702**. The auxiliary gear **704** engages with the frame gear **706**. The auxiliary gear **704** transmits the power of the fourth driving motor **702** to the frame gear **706** and performs a function of controlling the rotation speed

of the frame gear **706**.
(213) The frame gear **706** is mounted on the frame gear accommodating part **209** inside the housing **201**. The frame gear accommodating part **209** may be formed to correspond to the shape of the frame gear **706**. The frame gear **706** may be accommodated in the frame gear accommodating part **209**, and may rotate by the power of the fourth driving motor **702** received through the auxiliary gear **704**.
(214) Here, as the rack gear accommodating part **707** is formed on top of the frame gear **706**, when the frame gear **706** rotates as a whole, the rack gear accommodating part **707** and the rack gear **604** accommodated in the rack gear accommodating part **707** also rotate in response to the rotation of the frame gear **706**. Through this, the rack gear **604** may move from the first position to the second position as described above.
(215) In addition, a flat plate accommodating part **708** may be formed in the central portion of the frame gear **706**. The flat plate accommodating part **708** accommodates the aforementioned fixing portion **1005**, and as the fixing portion **1005** is coupled to the flat plate accommodating part **708**, the plate **1004** may rotate together in response to the rotation of the frame gear **706**.
(216) Similar to the relationship between the pressurizing part **404** (see FIGS. **13** to **17**) and the pressurizing opening part **512** (see FIGS. **13** to **17**) described above, the initial positions of the rack gear **604** and the plate opening **1006** (see FIG. **9**) may also correspond. In response to the rotation of the frame gear **706**, the rack gear **604** and the plate opening **1005** may also rotate together. Due to this, as the frame gear **706** rotates, the rack gear **604** and the flat plate opening **1006** may be positioned on at least one or more microneedle bodies **151** (see FIG. **3**).
(217) In summary, according to the operation of the fourth drive mechanism **700**, the rack gear **604** and the flat plate opening **1006** are positioned in the microneedle body **151** (see FIG. **3**). Thereafter, the rack gear **604** pressurizes the microneedle body **151** (see FIG. **3**) according to the operation of the third drive mechanism **600**. The microneedles formed in the pressurized microneedle body **151** (see FIG. **3**) protrude outward through the flat plate opening **1006**, so that the microneedles can be injected into the user's body.
(218) FIGS. **20** to **21** illustrate a drive mechanism according to another embodiment.
(219) FIG. **20**A and FIG. **20**B illustrate a drive mechanism according to yet another embodiment, and FIG. **21**A and FIG. **21**B illustrate a wedge-shaped microneedle patch used in the drive mechanism according to yet another embodiment.
(220) Referring to FIG. **20**A and FIG. **20**B, the applicator **200** (see FIG. **10**) according to an embodiment may be implemented through a fifth drive mechanism **800**.
(221) FIG. **20**A is a front view of the fifth drive mechanism, and FIG. **20**B is a bottom view of the fifth drive mechanism.
(222) The fifth drive mechanism **800** may include a fifth driving motor (not shown), a second main gear **804**, a second pressurizing part **806**, and a second lower plate **809**. The fifth drive mechanism **800** is preferably operated in interaction with the microneedle patch **900** having a wedge shape.
(223) The operation of the fifth drive mechanism **800** is almost similar to the operation of the second drive mechanism **500**. Since the operational relationship between the main gear **504** and the lower plate **510** may be applied to the operational relationship between the second main gear **804** and the second lower plate **808**, a detailed description thereof will be omitted.
(224) The fifth drive mechanism **800** may also include a second pressurizing part **806**. Unlike the first and second drive mechanisms **400** and **500**, the second pressurizing part **806** is coupled to the second shaft **850**. For this reason, the second pressurizing part **806** may rotate together with the second shaft **850** in response to the rotation of the second main gear **804**.
(225) A microneedle patch having a wedge shape will be described with reference to FIG. **21**A and FIG. **21**B.
(226) Referring to FIG. **21**A and FIG. **21**B, the microneedle patch **900** having a wedge shape may include a plurality of wedge bodies **901**. Here, the wedge body **901** may include an inclined surface

902 and a flat surface **904** on one surface. A microneedle **906** storing a drug may be formed on the opposite surface of each wedge body **901**. The microneedles **906** may be formed in a region corresponding to the flat surface **904**. The microneedle patch **900** may have a second shaft accommodating part **908** in which the second shaft **850** may be accommodated in the center thereof.
(227) A specific driving example of the fifth drive mechanism **800** will now be described with reference to FIGS. **20**A, **20**B, **21**A and **21**B.
(228) When the fifth driving motor (not shown) operates and the second main gear **804** rotates, the second lower plate **808** and the second pressurizing part **806** connected to the second main gear **804** through the second shaft **850** may rotate together.
(229) At this time, the microneedle patch **900** does not rotate, and the second pressurizing part **806** may move along the surface of the inclined surface **902**. As the second pressurizing part **806** moves along the inclined surface **902**, the microneedle patch **900** may be pressed downward. In this case, the second lower plate **808** may also be rotated, so that the second pressurizing opening part **809** may be located in the area where the microneedle **906** is formed.
(230) When the second pressurizing opening part **809** is positioned in the region where the microneedle **905** is formed, the second pressurizing part **806** moves on the flat surface **904**. When the second pressurizing part **806** moves on the flat surface **904** and pressurizes the wedge body **901**, a microneedle **906** located on the opposite surface of the flat surface **904** may protrude through the second v **809** to inject a drug to the user.
(231) After moving on the flat surface **904**, the second pressurizing part **806** may be positioned on another wedge body **901** formed in the microneedle patch **900**. Specifically, as the second main gear **804** further rotates, the second pressurizing part **806** may be located on the inclined surface **902** of the wedge body **901** different from the wedge body already pressurized. In this case, the second pressurizing opening part **809** may be located on the opposite surface of the microneedle patch **900** corresponding to the inclined surface **902** in which the microneedle **906** is not formed. Thus, it is possible to prevent foreign substances from entering the microneedle **906**.
(232) In the above, the drive mechanism of the applicator **200** according to various embodiments was described.
(233) Hereinafter, the overall operation of the applicator **200** according to the embodiment will be described, with the description focusing mainly on electrical control.
(234) FIG. **22** illustrates a control configuration of an applicator according to an embodiment.
(235) According to an embodiment, the applicator may include a controller **3000** and a memory **3002**. The controller **3000** and the memory **3002** may be implemented through a circuit design of the above-described PCB **203** (see FIG. **10**).
(236) The memory **3002** may store various types of information. Various data may be temporarily or semi-permanently stored in the memory **3002**. The memory **3002** may include, for example, a hard disk, SSD, flash memory, ROM, RAM, and the like. The memory **3002** may store an operating program for driving the applicator **200** or various data necessary for the operation of the applicator.
(237) For example, the memory **3002** may store structural information about the microneedle patch. Specifically, the memory **3002** may store information about the spacing between each microneedle body included in the microneedle patch. Also, the memory **3002** may store information on the positional relationship between each microneedle body included in the microneedle patch and the auxiliary region **151** (see FIG. **3**).
(238) As another example, the memory **3002** may store output information of the drive mechanism **210** (see FIG. **10**) corresponding to structural information about the microneedle patch. Specifically, the memory **3002** may include, as described above, output information of a drive mechanism for moving the pressurizing part **404** from the first position to the second position and output information of the drive mechanism for the pressurizing part **404** to pressurize the

microneedle patch.
(239) In addition to this, it will be apparent to those skilled in the art that all information for the operation of the applicator described above or to be described later in this specification may be stored in advance in the memory **3002**.
(240) The controller **3000** may control the overall operation of the applicator **200**. For example, the controller **3000** may control the operation of the drive mechanism **210** by transmitting a driving signal to the drive mechanism **210** (see FIG. **10**).
(241) The applicator **200** according to an embodiment may be set to inject the drug through the microneedle to the user at a predetermined time. The predetermined time may be set by the user or may be predetermined for a specific drug.
(242) FIG. **23** is a flowchart illustrating a method of controlling an applicator performed by a controller according to an embodiment.
(243) According to an embodiment, the control method of the applicator includes the steps of moving the pressurizing part from the first position to the second position S**100**, pressurizing the microneedle patch with the pressurizing part at the second position S**200**, returning the pressurizing part from the second position to the first position S**300**, moving the pressurizing part from the first position to the third position S**400**, and returning the pressurizing part from the third position to the first position S**500**.
(244) According to an embodiment, the controller **3000** may control the second drive mechanism **500** so that the pressurizing part **404** is positioned at the first position S**100**. Specifically, when the controller **3000** transmits a driving signal to the second drive mechanism **500**, the second drive mechanism **500** operates as much as a predetermined output, and due to this, as the main gear **504** rotates, the pressurizing part **404** may move from the first position to the second position.
(245) When the pressurizing part **404** moves from the first position to the second position, the controller **3000** may control the first drive mechanism **400** for the pressurizing part **404** to pressurize the microneedle patch S**200**. Specifically, when the controller **3000** sends a driving signal to the first drive mechanism **400**, the first drive mechanism may operate according to a predetermined output so that the pressurizing part **404** may pressurize the microneedle patch **100**.
(246) After the pressurizing part **404** pressurizes the microneedle patch **100**, the controller **3000** may control the second drive mechanism **500** so that the pressurizing part **404** is returned from the second position to the first position S**300**. Specifically, when the controller **3000** transmits a driving signal, the second drive mechanism operates by a predetermined output and, due to this, the main gear **504** rotates and accordingly, the pressurizing part **404** may be returned from the second position to the first position.
(247) After the pressurizing part **404** is returned to the first position, the controller **3000** may control the second drive mechanism **400** to move the pressurizing part **404** from the first position to the third position S**400**. Specifically, when the controller **3000** sends a drive signal to the second drive mechanism **400**, the second drive mechanism may be operated according to the predetermined output to move the pressurizing part **404** from the first position to the third position. When the pressurizing part **404** moves to the third position, the controller **3000** may control the first drive mechanism so that the pressurizing part **404** pressurizes the microneedle body corresponding to the third position.
(248) The microneedle patch **100** according to an embodiment is not allowed to be reused. Also, as described above, one microneedle body **151** may store one dose of drug. Therefore, the applicator **200** may inject the drug stored in one microneedle body to the user at the second position, and then inject the drug stored in the other microneedle body to the user at the third position. Thus, it will be apparent that the drug stored in the microneedle body present in the fourth position rather than the third position may be injected to the user thereafter.
(249) After the pressurizing part **404** pressurizes the microneedle body at the third position, the controller **300** may control the second drive mechanism **500** to return the pressurizing part **404**

from the third position to the first position.
(250) As such, as the controller **3000** controls the drive mechanism to return the pressurizing part **404** to the initial position (first position) after pressurizing the microneedle body at various positions, the pressurizing opening part **512**, which rotates corresponding to the pressurizing part **404**, is also returned to its initial position, when the applicator is in the standby state, the pressurizing opening part **512** may be located in the auxiliary region **140** (see FIG. **3**). Due to this, there is an effect that the contamination of the microneedle patch, which is vulnerable even to slight contamination, can be prevented.
(251) FIGS. **24** and **25** are views of the operating process of the applicator according to an embodiment from the back of the applicator.
(252) FIG. **24** illustrates the first drug injection operation of an applicator according to an embodiment viewed from the back of the applicator, and FIG. **25** illustrates the second drug injection operation of an applicator according to an embodiment viewed from the back of the applicator.
(253) Referring to FIG. **24**, in the standby state of the applicator, the pressurizing opening part **512** is placed at the first position. At this time, the pressurizing opening part **512** is placed in the auxiliary area **140**, so that the auxiliary area prevents foreign substances from entering from the outside.
(254) Thereafter, when the applicator operates, the pressing opening **512** may move from the first position to the second position. When the pressurizing opening part **512** moves from the first position to the second position, a body portion **151** of the microneedle patch is positioned in the pressurizing opening part **512**. Here, the microneedle may be exposed to the outside, and the microneedle may contact the user's body by the pressurization of the pressurizing part **404**.
(255) After injection of the drug, the pressurizing opening part **512** is returned to the first position again, and the applicator enters the standby state. Consequently, the auxiliary region **140** prevents foreign substances from being introduced from the outside and prevents the microneedle patch from being contaminated.
(256) Referring to FIG. **25**, a second drug injection operation is shown.
(257) When the applicator is operated from the standby state, the pressurizing opening part **512** may be moved from the first position to the third position instead of the second position. The third position may be expressed as a position corresponding to a microneedle body different from the microneedle body corresponding to the second position. When the drug injection at the third position is completed, the pressurizing opening part **512** may be returned to the initial position (the first position).
(258) As such, when a plurality of drug injections are performed for one microneedle patch, due to the drive mechanism, only drug injection into one microneedle body per one operation is performed, so that a fixed amount of drug injection is possible. Also, when the applicator enters the standby state after drug injection, the pressurizing opening part **512** is returned to its initial state to prevent contamination of the microneedle patch.
(259) In the above, the drug injection operation of the microneedle patch and the applicator was mainly examined.
(260) However, accurate injection of the drug may be performed when the applicator is properly mounted on the user's body and operated. To this end, the applicator according to an embodiment may further include a touch sensor for determining whether the applicator is correctly mounted on the user.
(261) FIG. **26** illustrates a touch sensor according to an embodiment.
(262) According to an embodiment, the applicator **200** may further include a touch sensor **1100**. The touch sensor **1100** may be attached to a lower cover **202**. Specifically, the touch sensor **1100** may be attached to the outside or the inside of the lower cover **202**. Also, the touch sensor **1100** may have a circular or annular shape corresponding to the shape of the lower cover **202**. In

addition, the touch sensor **1100** may be formed integrally with the lower cover **202**.
(263) For example, the touch sensor may be implemented as a capacitive touch sensor.
(264) The touch sensor **1100** may be divided into a plurality of zones. In the drawing, the touch sensor divided into first to fourth zones **1101**, **1102**, **1103**, and **1104** is illustrated by way of example. However, this is only an example, and it will be understood that a touch sensor divided into more or fewer zones may also be used.
(265) When an object contacts all or some areas **1101**, **1102**, **1103**, **1104** of the touch sensor **1100**, the amount of current flowing for each zone is changed, and due to this, the controller **3000** can determine whether the applicator is correctly mounted on the user's body.
(266) FIG. **27**A and FIG. **27**B are graphs illustrating an electrical signal measured by a touch sensor according to the mounting state of an applicator.
(267) The applicator may determine whether the applicator is properly mounted prior to actuating the drive mechanism. Specifically, the controller **3000** may determine whether the applicator is correctly mounted on the user's body based on an electrical signal value flowing through the touch sensor **1100**.
(268) In the graph shown in FIG. **27**A and FIG. **27**B, the x-axis means time. Units of measure can be see, ms, us, ns, etc., as well as all units of measure for time. It will be appreciated that the y-axis may be an electrical signal including a sensor output or a capacitance value, and any unit of measure representing them may be used. Although the measurement unit is not clearly disclosed in the graph of FIG. **27**A and FIG. **27**B, all measurement units representing time and electrical signals may be used.
(269) FIG. **27**A shows a case in which the applicator according to an embodiment is suitably mounted to the user's body. Referring to FIG. **27**A, it is shown that the amount of current measured in each region **1101**, **1102**, **1103**, **1104** of the touch sensor **1100** is equal to or greater than a threshold value.
(270) As such, when the value of the current measured in each zone of the touch sensor **1100** is equal to or greater than the threshold value, the controller **3000** may determine that the applicator is properly mounted on the user's body. If it is determined that the applicator is properly mounted, the controller **3000** may transmit a driving signal for drug injection to the drive mechanism. However, this is only an example and when the number of zones included in the touch sensor **1100** is large, even when a current greater than a threshold value flows in more than a predetermined number of zones, the controller **3000** may determine that the applicator is properly mounted on the user's body. It will be understood that various examples that may be modified in addition to this may be incorporated into the spirit of the present disclosure.
(271) FIG. **27**B shows a case in which the applicator according to an embodiment is not properly mounted on the user's body. Referring to FIG. **27**B, only the current value flowing in the first region **1101** included in the touch sensor **1100** was measured as a current value greater than or equal to the threshold value, and in the case of the remaining regions **1102**, **1103**, **1104**, the current values were measured as less than or equal to the threshold. In this case, the controller **3000** may determine that the applicator is not properly mounted on the user's body. If it is determined that the applicator is not properly mounted on the user's body, the controller **3000** may not transmit a driving signal for actuating the drive mechanism. Also, the controller **3000** may generate an indication signal reflecting that the applicator is not properly worn, and may provide it to an outside (user) through a display.
(272) In the above, the configuration and features of the present disclosure have been described based on the embodiments according to the present disclosure, however, the present disclosure is not limited thereto and it should be apparent to those skilled in the art to which the present disclosure pertains that various changes or modifications can be made within the spirit and scope of the present disclosure, and thus such changes or modifications fall within the scope of the appended claims.

Claims

1. A cartridge for accommodating a microneedle patch comprising: a body portion accommodated in an accommodating part formed in a housing of an applicator; a flat plate portion, mounted on the body portion, comprising a single flat plate opening through which a pressurizing part passes; and a needle body comprising a plurality of groups of microneedles spaced apart from each other, each microneedle group comprising a plurality microneedles and configured to pass through the single flat plate opening by movement of the pressurizing part, wherein the flat plate portion is rotatably mounted by including a fixing portion and is located below the microneedle patch to support the microneedle patch, and wherein the cartridge is configured to be independently inserted into and removed from the housing of the applicator so as to be replaceable.
2. The cartridge of claim 1, wherein the fixing portion is operatively coupled to a drive mechanism, and wherein the flat plate portion is configured to be rotated according to a driving of the drive mechanism.
3. The cartridge of claim 1, wherein an upper protective film is formed on an upper surface of the microneedle patch.
4. The cartridge of claim 1, wherein a lower protective film is formed on a lower surface of the microneedle patch and formed on an upper surface of the flat plate portion.
5. The cartridge of claim 1, wherein the flat plate opening is configured to enable the pressuring part to pass therethrough.
6. The cartridge of claim 1, wherein each microneedle group is configured to pass through the single flat plate opening once per rotation of the flat plate portion.
7. The cartridge of claim 1, wherein a single microneedle group of the plurality groups of microneedles is configured to pass through the single flat plate opening while the remaining microneedle groups do not pass through the single flat plate opening.